

Introduction to CSS (Cascading Style Sheets)

What is CSS?

CSS stands for **Cascading Style Sheets**. It is used to style and design web pages—such as changing text colors, fonts, layout, and more.

It separates the **structure** (HTML) from the **presentation** (CSS).

History

Why Was CSS Created?

In the early 1990s, web pages were made using only HTML. But HTML was meant for **structure**, not **design**. Developers started using tags like , <center>, and inline styles to style web content, which made the code:

- **Messy**
- **Hard to maintain**
- **Not reusable**

To solve this, **CSS was introduced to separate design from structure.**

Who Invented CSS?

- **Håkon Wium Lie**, a Norwegian web pioneer, proposed CSS in **October 1994**.
- He was working with **Tim Berners-Lee** (the inventor of HTML) at **CERN** and later at **W3C**.

- His idea was accepted and supported by the **World Wide Web Consortium (W3C)**.

Timeline of CSS Versions

CSS1 – Released in 1996

- First official version by W3C.
- Introduced basic styling features:
 - **Fonts, colors, text alignment**
 - **Margins, padding, borders**
- Limited support in early browsers (like Netscape Navigator and Internet Explorer 3).

CSS2 – Released in 1998

- Added more powerful features:
 - Positioning (absolute, relative, fixed)
 - Z-index, media types (screen, print)
 - More advanced selectors
- Better browser support came later (IE5+, Mozilla).

CSS2.1 – Revised in 2004–2011

- A cleaned-up version of CSS2.
- Removed features not well supported in browsers.
- Became a stable version that most modern browsers support.

CSS3 – Started around 2005 (Still evolving)

- Major upgrade with **modular design** (split into small parts like Flexbox, Grid, Animation, etc.).
- Introduced:
 - Rounded corners (border-radius)
 - Shadows (box-shadow, text-shadow)
 - Transitions and animations
 - Media queries (for responsive design)
 - Flexbox, Grid Layout

CSS4 – Not officially released

- There is no full CSS4 version.
- CSS continues to evolve through modules (e.g., Selectors Level 4, Color Level 4).
- So, “**CSS4**” is just a collection of newer modules, not a real version.

Modern CSS Today

- Widely supported in **all modern browsers** (Chrome, Firefox, Safari, Edge).
- Powerful features for:
 - **Responsive Design**
 - **Dark Mode**
 - **Animations and Transitions**
 - **Frameworks** like Bootstrap, Tailwind

Summary

Version	Year	Key Features
CSS1	1996	Basic styling: fonts, colors, margins
CSS2	1998	Positioning, z-index, media types
CSS2.1	2004–2011	Browser-compatible cleanup
CSS3	2005+	Modules, animations, flexbox, media queries
CSS4	–	Not a real version, modular evolution

Why Use CSS?

- To apply **consistent styling** across multiple HTML pages.
- Makes websites more **attractive** and **user-friendly**.
- Enhances **layout control** and **responsiveness**.

Types of CSS

Type	Description	Example
Inline CSS	Style is added directly to the HTML tag	<code><p style="color: red;">Text</p></code>
Internal CSS	Style is written inside <code><style></code> tag in <code><head></code> section	<code><style>p {color: red;}</style></code>
External CSS	Style is written in a separate .css file and linked to HTML	<code><link rel="stylesheet" href="style.css"></code>

CSS Syntax

```
selector {  
  property: value;  
}
```

Example:

```
h1 {  
  color: blue;  
  font-size: 24px;  
}
```

Common CSS Properties

Property	Description	Example
color	Text color	color: red;
background-color	Background color	background-color: yellow;
font-size	Size of text	font-size: 20px;
font-family	Font style	font-family: Arial;
text-align	Aligns text	text-align: center;
margin	Space outside an element	margin: 10px;
padding	Space inside an element	padding: 15px;
border	Border around elements	border: 1px solid black;

Selectors in CSS

Selector	Description	Example
*	Selects all elements	* {margin: 0;}
element	Selects specific tags	p {color: green;}
.class	Selects elements with a specific class	.title {font-size: 24px;}
#id	Selects element with a specific ID	#header {background: gray;}
element, element	Group selectors	h1, h2 {color: red;}

CSS Comments

/* This is a comment */

Example: Internal CSS Code

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
  body {
```

```
    background-color: lightblue;
```

```
  }
```

```
  h1 {
```

```
    color: navy;
    text-align: center;
}
p {
    font-size: 18px;
}
</style>
</head>
<body>
    <h1>Welcome to CSS</h1>
    <p>This is a styled paragraph.</p>
</body>
</html>
```

Cascading Order (Priority of CSS)

1. **Inline CSS** – highest priority
2. **Internal CSS**
3. **External CSS** – lowest priority

CSS File Linking It to HTML

Types of CSS

1. **Inline CSS:** Inside an HTML element (not recommended for large projects)
2. **Internal CSS:** Inside <style> in the HTML file
3. **External CSS:** In a separate .css file (recommended for modularity)

Example of Inline CSS in HTML

Inline CSS is used to apply a unique style to a single HTML element. It is added directly in the style attribute of the element.

Example:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Inline CSS Example</title>
```

```
</head>
```

```
<body>
```

```
<h1 style="color: blue; text-align: center;">Welcome to Inline  
CSS</h1>
```

```
<p style="font-size: 18px; color: green;">
```

This paragraph is styled using inline CSS.

</p>

<button style="background-color: orange; color: white; padding: 10px;">

Click Me

</button>

</body>

</html>

Internal CSS

Internal CSS is used to define styles in the <head> section of an HTML document using the <style> tag. It applies styles to the entire page or specific elements.

Example:

<!DOCTYPE html>

<html>

<head>

<title>Internal CSS Example</title>

<style>

body {

background-color: #f0f8ff;

font-family: Arial, sans-serif;

```
}
```

```
h1 {  
  color: navy;  
  text-align: center;  
}
```

```
p {  
  font-size: 18px;  
  color: darkgreen;  
}
```

```
button {  
  background-color: darkorange;  
  color: white;  
  padding: 10px 15px;  
  border: none;  
  border-radius: 5px;  
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1>Welcome to Internal CSS</h1>
```

`<p>This paragraph is styled using internal CSS. </p>`

`<button>Click Me</button>`

`</body>`

`</html>`

External CSS

External CSS is written in a separate .css file and linked to the HTML file using the `<link>` tag. It is ideal for styling multiple HTML pages consistently.

Step 1: Create the HTML File (index.html)

`<!DOCTYPE html>`

`<html>`

`<head>`

`<title>External CSS Example</title>`

`<link rel="stylesheet" type="text/css" href="style.css">`

`</head>`

`<body>`

`<h1>Welcome to External CSS</h1>`

`<p>This paragraph is styled using external CSS. </p>`

`<button>Click Me</button>`

`</body>`

`</html>`

Step 2: Create the CSS File (style.css)

```
body {  
  background-color: #e0f7fa;  
  font-family: Verdana, sans-serif;  
}
```

```
h1 {  
  color: #006064;  
  text-align: center;  
}
```

```
p {  
  font-size: 18px;  
  color: #00796b;  
}
```

```
button {  
  background-color: #ff9800;  
  color: white;  
  padding: 10px 15px;  
  border: none;  
  border-radius: 5px;  
}
```

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Notes on CSS Selectors

CSS selectors are used to **target HTML elements** that you want to style. They tell the browser **which elements** the CSS rules apply to.

Types of CSS Selectors

1. Universal Selector (*)

- Selects **all elements** on the page.

```
* {  
margin: 0;  
padding: 0;  
}
```

2. Element Selector (Tag Name)

- Selects all elements of a **specific type**.

```
p {  
color: green;  
font-size: 16px;  
}
```

3. Class Selector (.)

- Selects all elements with a **specific class**.

```
. myClass {  
    background-color: yellow;  
    font-weight: bold;  
}
```

In HTML:

```
<p class="myClass">Styled using class selector</p>
```

4. ID Selector (#)

- Selects a single element with a **specific ID (unique)**.

```
#mainHeading {  
    color: blue;  
    text-align: center;  
}
```

In HTML:

```
<h1 id="mainHeading">Welcome</h1>
```

5. Group Selector (,)

- Applies the same style to **multiple elements**.

```
h1, h2, p {  
    font-family: Arial;  
}
```

6. Descendant Selector (space)

- Selects elements that are **inside** another element.

```
div p {  
  color: red;  
}
```

7. Child Selector (>)

- Selects **direct child** elements.

```
ul > li {  
  list-style-type: square;  
}
```

8. Adjacent Sibling Selector (+)

- Selects the **next immediate sibling**.

```
h1 + p {  
  font-style: italic;  
}
```

9. Attribute Selector

- Selects elements with a specific attribute.

```
input[type="text"] {  
  border: 1px solid #ccc;  
}
```


10. Attribute Selector

- Selects elements by their attribute name or value.

```
input[type="text"] {  
  border: 2px solid blue;  
}
```

11. Pseudo-Class Selector Examples (:)

Pseudo-classes style elements **based on state or position**.

Example 1:

```
<a href="#">Hover Over Me</a>  
a:hover {  
  color: red;  
  text-decoration: underline;  
}
```

When you hover over the link, the color changes to red.

Example 2: :first-child

```
<ul>  
  <li>First Item</li>  
  <li>Second Item</li>  
</ul>
```

```
li:first-child {  
  color: blue;  
}
```

Only the first will be blue.

Example 3: :nth-child(2)

```
<p>Paragraph 1</p>  
<p>Paragraph 2</p>  
<p>Paragraph 3</p>  
p:nth-child(2) {  
  color: green;  
}
```

Only the **second** <p> will turn green.

Example 4: :checked

```
<input type="checkbox" id="agree">  
<label for="agree">I agree</label>  
input:checked + label {  
  color: green;  
  font-weight: bold;  
}
```

When the checkbox is checked, the label text becomes bold and green.

12. Pseudo-Element Selector Examples (::)

Pseudo-elements style **specific parts** of an element.

Example 1:

```
<h2>Welcome</h2>
```

```
h2::before {  
  content: "👉";  
  color: orange;  
}
```

Displays 👉 before the word **Welcome**.

Example 2:

```
<h2>Start</h2>
```

```
h2::after {  
  content: "☑";  
  color: green;  
}
```

Displays ☑ after the word **Start**.

Example 3:

<p>This is a long paragraph that contains multiple lines of text.</p>

```
p::first-line {  
  font-weight: bold;  
  color: red;  
}
```

Only the **first line** of the paragraph appears in bold red.

Example 4: ::selection

```
::selection {  
  background-color: yellow;  
  color: black;  
}
```

Pseudo-Class Selector (:

- Selects elements in a specific state.

Selector	Description	Example
:hover	When mouse is over	a:hover
:first-child	First child of a parent	p:first-child
:last-child	Last child of a parent	li:last-child
:nth-child(n)	nth child	tr:nth-child(2)
:checked	Checked input	input:checked

Pseudo-Element Selector (::)

- Targets part of an element.

Selector	Description	Example
::before	Insert content before element	<code>h1::before { content: "* "; }</code>
::after	Insert content after element	<code>h1::after { content: " ✓"; }</code>
::first-line	Style first line of text	<code>p::first-line { font-weight: bold; }</code>

Summary Table

Selector Type	Example	Selects
Universal	*	All elements
Element	p	All <p> tags
ID	#title	Element with id="title"
Class	.menu	Elements with class="menu"
Grouping	h1, p	All <h1> and <p>
Descendant	div p	All <p> inside <div>
Child	ul > li	Direct children of
Adjacent	h1 + p	First <p> after <h1>
Sibling	h2 ~ p	All <p> after <h2>
Attribute	input[type="text"]	Text input fields
Pseudo-class	a:hover	Hovered links
Pseudo-element	p::first-line	First line of paragraph

Summary Table:

Selector Type	Symbol / Syntax	Example
Universal	*	* {}
Element	tagname	p {}
Class	. classname	. box {}
ID	#idname	#header {}
Group	selector, selector	h1, p {}
Descendant	ancestor descendant	div p {}
Child	parent > child	ul > li {}
Adjacent Sibling	element + element	h1 + p {}
Attribute	[attr=value]	input[type="text"]

Example:

```
<!DOCTYPE html>

<html>

<head>

  <title>All CSS Selectors Example</title>

  <style>

    /* 1. Universal Selector */
    * {
      font-family: Arial, sans-serif;
    }

    /* 2. Element Selector */
    h1 {
      color: darkblue;
      text-align: center;
    }

    /* 3. Class Selector */
    .highlight {
      background-color: yellow;
      font-weight: bold;
    }

    /* 4. ID Selector */
```

```
#special {  
    color: green;  
    font-size: 20px;  
}
```

```
/* 5. Group Selector */  
h2, p {  
    margin: 10px;  
}
```

```
/* 6. Descendant Selector */  
div p {  
    color: red;  
}
```

```
/* 7. Child Selector */  
ul > li {  
    color: blue;  
}
```

```
/* 8. Adjacent Sibling Selector */  
h2 + p {  
    font-style: italic;  
}
```

```
/* 9. Attribute Selector */
input[type="text"] {
    border: 2px solid orange;
    padding: 5px;
}
</style>
</head>
<body>

<h1>CSS Selectors Example</h1>

<!-- Element Selector -->
<h2>Subheading (Group Selector) </h2>

<!-- Adjacent Sibling Selector will apply here -->
<p>This paragraph is styled after h2 (Adjacent Sibling) </p>

<!-- Class Selector -->
<p class="highlight">This is a highlighted paragraph (Class Selector)
</p>

<!-- ID Selector -->
<p id="special">This paragraph has a unique ID (ID Selector) </p>
```

<! -- Descendant Selector -->

<div>

<p>This paragraph is inside a div (Descendant Selector) </p>

</div>

<! -- Child Selector -->

First Item (Child Selector)

Second Item

<! -- Attribute Selector -->

<label>Enter Name:</label>

<input type="text" placeholder="Your name">

</body>

</html>